Supplementary material

lit-acquisition: multi-source, multilingual biomedical literature acquisition with citation-graph traversal

1 Reference set: construction and what it can support

The reference set contains 43 clinically actionable variants and 822 papers. Construction followed a documented automated protocol: for each variant a clinical-style natural-language query was formulated, candidates were pooled from PubMed E-utilities, Europe PMC and Crossref, and a weighted heuristic scored each candidate (gene symbol in title +3; variant identifier +5; disease keywords up to +3; case-report publication type +2; reviews −2; editorials, letters, comments and news excluded). The 30 original variants were annotated in two rounds; 13 further variants were added on 2026-08-14 by `benchmarks/gold_standard/build_variants.py`, which replicates the same protocol without modifying the originals.

Limitations that bound every claim in the paper. (i) The annotations are *heuristic and not expert-verified*, so they contain unquantified false positives and false negatives. (ii) The candidate pool was drawn from channels that *overlap the evaluated system*, a pooling/circularity bias: absolute recall against this set measures agreement with the annotation protocol, not clinical completeness. (iii) Language labels come from provider metadata; 305 of 822 papers have no label, and only German (13) and French (11) carry non-English annotations, so claims about other target languages are not supported by this set. Consequently all reported metrics are *relative* coverage comparisons. `benchmarks/gold_standard/variants.jsonl` is the source of truth; `variants_annotated.jsonl` is a historical V001–V030 file.

2 Configurations and evaluation protocol

Four configurations were compared at a uniform budget of 100 keyword candidates per variant:

- **M1 PubMed-only** — PubMed E-utilities only, English query.
- **M2 multi-source** — all channels, English query.
- **M3 +translation** — adds LLM translation into en/zh/ja/de/fr/ru with language-aware routing.
- **M4n +citation (nested)** — strictly nested ablation built from the frozen M3 ranked list plus citation-graph items recorded in the independent M4 run, deduplicated and appended in traversal order (monotonicity verified; the keyword phase is not re-executed).

Metrics are macro means over the 43 variants: SetRecall (fraction of a variant's reference papers retrieved), Recall@K, Precision@K (fixed denominator K), Precision@Retrieved, and nDCG@K for $K \in \{5, 10, 15, 20, 50\}$. Micro aggregates are also reported in the artifacts. Statistics: two-sided paired Wilcoxon signed-rank tests with zeros dropped, Holm–Bonferroni correction over a pre-declared family of eight tests, matched-pairs rank-biserial effect size, and 10,000 variant-level

percentile bootstrap resamples (seed 42). Live-API run-to-run variance was estimated from two independent executions of the M3 configuration (18/30 identical ranked lists; mean Jaccard 0.891).

Reproducibility of the LLM components. The recorded collection run used `Qwen/Qwen2.5-72B-Instruct` for translation (temperature 0.2, max_tokens 1024, system-prompt SHA-256 prefix `96fe414ab84606dd`) and `BAAI/bge-reranker-v2-m3` as the cross-encoder rerank stage. All of these are recorded in `benchmarks/results/run_manifest.json`. The relevance gate was *disabled* during collection, so the reported metrics are raw retrieval comparisons; the lite edition performs no LLM calls at all.

3 Full metrics

Table 1: Full per-method metrics over 43 variants (macro means). Precision@K uses a fixed denominator K; Precision@Retr. is over the retrieved list.

Method	SetRec.	Rec@5	Rec@20	Rec@50	P@20	nDCG@20	nDCG@50
M1 PubMed-only	0.269	0.076	0.171	0.218	0.155	0.205	0.219
M2 multi-source	0.472	0.044	0.153	0.275	0.147	0.164	0.222
M3 +translation	0.465	0.085	0.276	0.446	0.265	0.301	0.381
M4n +citation (nested)	0.467	0.085	0.276	0.446	0.265	0.301	0.381

4 Significance tests

Table 2: Pre-declared eight-test family: paired Wilcoxon signed-rank with Holm–Bonferroni correction, 10,000 bootstrap resamples (seed 42). r is the matched-pairs rank-biserial correlation.

Contrast	Δ	95% CI	p_{Holm}	r
M2 vs M1 set recall	+0.203	[+0.128, +0.277]	1.08e-04	+0.77
M3 vs M2 set recall	-0.007	[-0.087, +0.069]	1.00e+00	+0.02
M4n vs M3 set recall	+0.002	[+0.000, +0.004]	1.00e+00	+1.00
M4n vs M1 set recall	+0.198	[+0.123, +0.269]	1.93e-04	+0.74
M4n vs M2 set recall	-0.005	[-0.086, +0.071]	1.00e+00	+0.03
M4n vs M1 recall20	+0.105	[+0.035, +0.170]	1.61e-02	+0.51
M4n vs M1 ndcg50	+0.162	[+0.086, +0.234]	4.30e-04	+0.66
M4n vs M3 prec retr	-0.011	[-0.016, -0.006]	1.91e-06	-1.00

5 Comparison with single-channel baselines

Note that the four baselines that only reorder the shared PubMed candidate pool (mean 80.7 candidates: dense, cross-encoder, agentic and multilingual dense) *cannot* exceed M1 on set recall by construction — their SetRecall columns are identical to M1's. The informative comparison is the hybrid BM25+dense+cross-encoder baseline.

Table 3: Single-channel baselines (uniform 20-paper budget over the same PubMed candidate pool unless noted). Significance is versus M4n.

Method	SetRecall	Recall@20	p_{Holm} vs M4n
S1 Dense (bge-m3)	0.269	0.182	3.86e-04
S2 Cross-encoder rerank	0.269	0.171	3.86e-04
S3 Hybrid BM25+dense+rerank	0.421	0.271	1.00e+00
S4 LLM query expansion	0.270	0.173	3.86e-04
S5 Agentic 2-round search	0.269	0.171	3.86e-04
S6 Multilingual dense (bge-m3)	0.269	0.181	3.86e-04
M4n pipeline	**0.467**	**0.276**	—

6 Pooled reference-set expansion (E2c)

The frozen reference set was built by scoring a capped 2026-08-14 pool with the documented rule (gene +3, variant +5, disease +1, threshold > 4). Re-applying that same rule to the pooled candidate lists of all ten recorded systems (11,161 candidate papers) judges 313 further papers relevant on 15 variants, enlarging the reference set from 822 to 1135 papers. Every system contributes to the pool, so no single system is favoured; the judgments are title-based, exactly like the frozen set. Table 4 shows that the manuscript's headline contrasts survive the expansion.

Table 4: Reference-set expansion: macro metrics on the frozen 822-paper set and on the pooled expansion (1135 papers).

	SetRecall		Recall@20		nDCG@20	
Method	frozen	pooled	frozen	pooled	frozen	pooled
M1 PubMed only	0.269	0.262	0.171	0.147	0.205	0.223
M2 multi-source	0.472	0.505	0.153	0.153	0.164	0.217
M3 +translation	0.465	0.445	0.276	0.267	0.301	0.368
M4n nested full pipeline	0.467	0.447	0.276	0.267	0.301	0.368
S3 hybrid (recorded)	0.421	0.416	0.271	0.254	0.332	0.393
S1 dense (recorded)	0.269	0.262	0.182	0.183	0.229	0.301

Table 5: Headline paired tests on the pooled expansion (Holm-corrected within the declared family).

Contrast	Δ	p_{Holm}	r	significant
M4n vs M1, SetRecall	+0.184	2.7×10^{-4}	+0.73	yes
M2 vs M1, SetRecall	+0.242	$< 10^{-4}$	+0.87	yes
M4n vs M1, Recall@20	+0.120	0.001	+0.63	yes
M4n vs M1, nDCG@20	+0.146	9.7×10^{-4}	+0.62	yes
M4n vs S3, SetRecall	+0.031	0.586	+0.10	no

7 Deconfounding the M2–M3 contrast (E1)

The recorded M2 order is the federation's completion order (no dedup, no ranking, no cross-encoder), while M3 runs the full workflow, so the published M2→M3 gain mixed ranking with translation. Rebuilding both pools through the *same* code path turns the comparison into a

symmetric 2×2 factorial (translation $\times$ ranking) over the frozen candidate lists. Table 6 reports the cells; Table 7 the Holm-corrected paired tests.

Table 6: Symmetric 2×2 factorial (translation $\times$ ranking) on the frozen pools; macro means over 43 variants. Within a row the pools are identical, so differences are due to the ranking stage alone.

Method	Recall@20	nDCG@20	Prec@Retr	SetRecall
M1 PubMed only (recorded)	0.171	0.205	0.145	0.269
M2 recorded federation order	0.153	0.164	0.087	0.472
M2_lex dedup + lexical rank	0.218	0.256	0.087	0.472
M2_rerank + cross-encoder	0.152	0.163	0.087	0.472
M3 recorded order (manuscript)	0.276	0.301	0.132	0.465
M3_lex dedup + lexical rank	0.306	0.342	0.132	0.465
M3_rerank + cross-encoder	0.279	0.302	0.132	0.465
M4n nested full pipeline	0.276	0.301	0.121	0.467

Table 7: Declared paired-test family for E1 (frozen reference set, Holm-corrected).

Test	Δ	p_{Holm}	r	significant
`ranking_on_multisource_recall20`	$+0.065$	9.9×10^{-4}	$+0.78$	yes
`crossencoder_on_multisource_recall20`	-0.066	9.9×10^{-4}	-0.80	yes
`translation_lexical_rung_recall20`	$+0.088$	9.9×10^{-4}	$+0.75$	yes
`translation_reranked_rung_recall20`	$+0.127$	$< 10^{-4}$	$+0.94$	yes
`crossencoder_on_translated_recall20`	-0.027	0.182	-0.47	no
`multisource_vs_m1_setrecall`	$+0.203$	1.4×10^{-4}	$+0.77$	yes
`multisource_vs_m1_recall20`	-0.020	1.000	$+0.00$	no
`translation_lexical_rung_setrecall`	-0.007	1.000	$+0.02$	no
`translation_reranked_rung_setrecall`	-0.007	1.000	$+0.02$	no
`raw_federation_vs_m1_setrecall`	$+0.203$	1.4×10^{-4}	$+0.77$	yes

Three findings matter for the manuscript. (i) Ranking is a real effect: dedup + lexical ranking lifts Recall@20 from 0.153 to 0.218 on the multi-source pool. (ii) Translation is the dominant shallow-ranking effect and it is now measured with the ranking stage held fixed: $+0.088$ Recall@20 at the lexical rung and $+0.127$ at the reranked rung, with SetRecall unchanged (-0.007), i.e. translation changes *which* papers are reachable rather than how many (German 11/13, French 10/11). (iii) The title-only cross-encoder does not help this reference set: -0.066 Recall@20 on the multi-source pool (Holm-significant) and -0.027 on the translated pool (not significant after correction); the interaction term is $+0.039$. Because reference-set membership itself requires an explicit variant mention, exact-match lexical ranking is structurally advantaged; the ClinVar comparison below provides the independent check. One artifact caveat is reported rather than silently fixed: the recorded M3 permutation is not reproducible by re-applying the documented ranker to the stored M3 list (Recall@20 0.276 recorded vs 0.306 re-ranked), so both orders are reported and the recorded one is kept for continuity with the manuscript's table.

8 Offline language backfill (E4a)

305 of 822 reference papers had no provider language label. A deterministic classifier (Unicode script detection plus weighted function/domain vocabulary and orthographic markers; English as the competing hypothesis) recovers a label for 300 of them, leaving 5 undecided. Thresholds

were selected on a calibration half of the provider-labelled papers and are reported on a held-out half: accuracy 0.984, non-English precision 0.933 and recall 0.933. The audit also exposes 8 provider labels contradicted by evidence (non-Latin-script titles, or journals written in the detected language), e.g.
five Chinese articles labelled `en`. Backfilling raises the German bucket from 13 to 18 papers and the French from 11 to 20.

Table 8: German/French reference papers recovered, before and after the offline language backfill. The multilingual claim does not depend on the 305 unlabelled papers: it strengthens as the non-English denominator grows.

	German		French	
Method	before	after	before	after
M1 PubMed only	0/13	0/18	0/11	0/20
M2 multi-source	1/13	1/18	0/11	0/20
M3 +translation	11/13	16/18	10/11	18/20
M4n nested full pipeline	11/13	16/18	10/11	18/20

9 Independent truth: ClinVar submitter citations (E2a)

To remove this project from the truth loop we collected the literature that clinical laboratories cite when they submit a variant interpretation to ClinVar (`esearch`/`esummary` on ClinVar, `elink` to PubMed, generic guidelines removed by a cross-variant filter). Records were found for 36 of 43 variants, yielding 3,275 curated citations. The systems' candidate pools contain 358 of the 3,275 (variant, paper) matches, so absolute recall is bounded by pool reach rather than by the reference set.

Table 9: Coverage of ClinVar-curated citations by the frozen candidate lists (36 variants with records; macro means, "found" counts (variant, paper) matches).

Method	SetRecall	Recall@20	Recall@50	Found
M1 PubMed only	0.045	0.016	0.027	143
M2 multi-source	0.064	0.016	0.032	180
M3 +translation	0.016	0.012	0.015	53
M4n nested full pipeline	0.021	0.012	0.015	65
S1 dense (recorded)	0.045	0.027	0.037	143
S3 hybrid (recorded)	0.080	0.036	0.056	237
S6 multilingual dense (recorded)	0.045	0.023	0.038	143

The result is a negative one, and it is reported as such: on this independent truth the pipeline does *not* outperform PubMed-only, and the federated multi-source configuration does slightly better than M4n. The reason is temporal, not methodological: the curated citations have median year 2011 (34

10 Unbound-pool SOTA comparison (E3)

Four of the six recorded baselines only reorder the PubMed pool, so their set recall equals M1's by construction, and the recorded S2/S3 cross-encoder stage was inert (see the defect note below). Applying every ranker through one cached code path on the frozen pools separates the two questions:

Table 10: Paired tests against the ClinVar truth (Holm-corrected).

Test	Δ	p_{Holm}	r	significant
`m4n_vs_m1_setrecall`	−0.0239	0.308	−0.46	no
`m4n_vs_m1_recall20`	−0.0041	1.000	−0.16	no
`m2_vs_m1_setrecall`	+0.0190	0.525	+0.39	no
`m3_vs_m1_setrecall`	−0.0293	0.022	−0.71	yes
`translation_lexical_recall20`	−0.0006	1.000	+0.05	no
`translation_reranked_recall20`	−0.0041	0.590	−0.41	no
`ranking_lexical_recall20`	−0.0036	0.519	−0.48	no
`crossencoder_recall20`	+0.0033	0.525	+0.47	no
`crossencoder_translated_recall20`	−0.0002	1.000	−0.07	no
`m4n_vs_s3_setrecall`	−0.0590	2.4×10^{-4}	−0.93	yes

Table 11 shows that within a track the candidate set — and therefore SetRecall — is identical, so only the ranking metrics move.

Table 11: Dual-track comparison on frozen pools: Track 1 fixes the PubMed pool (the M1 pool), Track 2 the federated pool (the M2 pool). Every ranker is applied by the same cached implementation; SetRecall is constant within a track.

Method	Recall@20	nDCG@20	Prec@20	SetRecall
T1 recorded PubMed order (=M1)	0.171	0.205	0.155	0.269
T1 + pipeline lexical rank	0.205	0.259	0.188	0.269
T1 + bge-m3 dense (S1 recipe)	0.181	0.228	0.162	0.269
T1 + cross-encoder (S2 recipe, corrected)	0.175	0.206	0.157	0.269
T1 + RRF(dense, order) (S3 fusion stage)	0.225	0.277	0.205	0.269
T1 + multilingual dense (S6 recipe)	0.180	0.224	0.162	0.269
T2 recorded federation order (=M2)	0.153	0.164	0.147	0.472
T2 + pipeline lexical rank	0.218	0.256	0.207	0.472
T2 + cross-encoder	0.152	0.163	0.145	0.472
T2 + bge-m3 dense (S1 recipe)	0.172	0.188	0.166	0.472
T2 + cross-encoder (S2 recipe, corrected)	0.152	0.163	0.145	0.472
T2 + RRF(dense, order) (S3 fusion stage)	0.165	0.184	0.159	0.472
T2 + multilingual dense (S6 recipe)	0.172	0.181	0.163	0.472
M3 recorded (federated + translation)	0.276	0.301	0.265	0.465
M4n nested full pipeline	0.276	0.301	0.265	0.467

Two conclusions follow. Federation, not ranking, produces the set-level gain: within each track SetRecall is fixed, while across tracks it moves from 0.269 to 0.472. And on a single pool no modern ranker beats the pipeline's own lexical ranking of the federated pool — dense, cross-encoder and multilingual-dense rankers land between the recorded order and lexical ranking on the reference set — which is why the headline comparison in the manuscript is a set-coverage comparison and not a reranking claim.

11 Reranking: a helper defect, corrected measurements and gating (E4c)

While building the deconfounding experiment we found that the shared benchmark helper `rerank_docs` restored the provider response to *input* order (`sorted(..., key=index)`) and its callers then truncated with `[:limit]`. The recorded S2 (cross-encoder) and S3 (hybrid) baselines and the

Table 12: Declared ranker comparisons within each track (Recall@20, Holm-corrected).

Test	Δ	p_{Holm}	r	significant
`t1_lexical_vs_recorded`	+0.0333	1.000	+0.33	no
`t1_dense_vs_recorded`	+0.0094	1.000	+0.19	no
`t1_rerank_vs_recorded`	+0.0033	1.000	+0.08	no
`t1_hybrid_vs_recorded`	+0.0532	0.016	+0.76	yes
`t1_mdense_vs_recorded`	+0.0086	1.000	+0.19	no
`t1_hybrid_vs_dense`	+0.0439	0.023	+0.78	yes
`t2_lexical_vs_recorded`	+0.0651	0.002	+0.78	yes
`t2_dense_vs_recorded`	+0.0195	0.790	+0.36	no
`t2_rerank_vs_recorded`	−0.0010	1.000	−1.00	no
`t2_hybrid_vs_recorded`	+0.0118	1.000	+0.33	no
`t2_mdense_vs_recorded`	+0.0189	1.000	+0.26	no
`t2_lexical_vs_dense`	+0.0457	0.229	+0.45	no

`rerank_eval.json` optimization artifact therefore contain *no* cross-encoder effect: their order is the unranked pool order. The helper is fixed (regression test `tests/test_rerank_ordering.py`); live re-runs of the affected baselines were not necessary because the frozen pools allow every ranker to be re-applied offline through one cached, corrected code path. With the corrected ordering, re-ranking the nested pipeline list (M5) leaves the shallow metrics where they were (Recall@20 +0.0001, nDCG@20 −0.0010, both n.s.), whereas the citation gate (M5g: keep only citation items whose rerank score reaches the minimum score of the top-20 M3 candidates) removes the precision cost of citation traversal without losing its two relevant papers: Precision@Retrieved +0.0103 ($p_{\text{Holm}} < 10^{-4}$), SetRecall −0.0016 (n.s.), keeping 12 of 299 citation items. The gate is the recommended operating point when precision matters.

Table 13: Reranking and citation gating with the corrected cross-encoder ordering (frozen reference set); `rerank_eval_corrected.json` supersedes `rerank_eval.json`.

Method	Recall@20	nDCG@20	Prec@Retr	SetRecall
M4n nested (recorded)	0.276	0.301	0.121	0.467
M5 full rerank (corrected)	0.276	0.300	0.121	0.467
M5g gated citation (corrected)	0.276	0.301	0.131	0.465

Widening the citation seed from one to five top-ranked M3 DOIs is not a free lunch: k=1 appends 151 citation papers and contributes 2 relevant ones (pooled reference set), k=3 appends 400 for 3, and k=5 appends 624 for 6, while Precision@Retrieved falls significantly at every step (−0.0095 and −0.0153 respectively, both $p_{\text{Holm}} < 0.001$). The single-seed default with the gate above is the better operating point.

12 Relevance-gate measurement (E4b)

The benchmark ran with `relevance_gate=False`, so the gate advertised in Figure 1 had no measured numbers. We ran it (production prompt) over a deterministic, variant-stratified sample of 120 candidate papers (60 reference-set papers and as many protocol-rejected candidates, so the sample prior is 50/50 and only the *rates* are interpretable), feeding PubMed abstracts through its `parsed_markdown` input. The production default model (`gpt-5` at `linxi.chat`) was rate-limited (HTTP 429) throughout the measurement window, so the identical prompt was run with

`Qwen/Qwen2.5-72B-Instruct`; the substitution is recorded in the artifact and the measurement should be repeated on the production model. The gate retained 96.7

13 Double annotation package (E2b, status: awaiting raters)

The expert re-annotation called for in the paper is prepared as a ready-to-run package: a deterministic, disease-stratified sample of 15 variants (V003, V006, V012, V015, V017, V019, V020, V022, V025, V027, V029, V032, V036, V037, V042) and 225 papers (150 heuristic positives, 75 heuristic negatives drawn from the pooled candidates the protocol rejected), two identical blank sheets with the Level 1 (topic relevance) and Level 2 (variant-level evidence) criteria, and an analysis script that reports Cohen's κ per level, the heuristic's precision/recall against each rater and the consensus, and the list of disagreements for adjudication. Files: `benchmarks/annotation/` (`e2b_sample.csv`, `e2b_sheet_raterA.csv`, `e2b_sheet_raterB.csv`, `e2b_key.csv`) and `benchmarks/analysis/e2b_agreement.py`. The target is $\kappa \geq 0.70$; the resulting numbers will replace the provisional truth used by the ClinVar and relevance-gate analyses.

14 Multilingual recovery

Table 14: Annotated German/French reference papers recovered per method. Only these two languages carry non-English annotations.

Method	German (of 13)	French (of 11)	Total (of 24)
M1 PubMed-only	0	0	0
M2 multi-source	1	0	1
M3 +translation	11	10	21
M4n +citation (nested)	11	10	21
Best English-only baseline (S3 hybrid)	1	0	1

Translation is set-level neutral in aggregate (M3 vs M2: $p_{\text{Holm}} = 1.0$) but changes which papers are reachable, which is the practical argument for language-aware routing.

15 Citation-graph exclusivity

Across all 43 variants, citation traversal contributed exactly two relevant reference papers, one of which appears in no keyword list:

- V002 (BRCA1 c.5266dupC) — *Familial history and prevalence of BRCA1, BRCA2 and TP53 pathogenic variants in HBOC Brazilian patients* (Sci Rep 2022, `10.1038/s41598-022-23012-3`).
- V035 (HFE p.C282Y) — *HFE p.C282Y Polymorphism and Risk of Metabolic Syndrome Components: Systematic Review and Meta-Analysis* (Medicina 2026, `10.3390/medicina62030589`).

The same step costs list precision (Precision@Retrieved -0.011, $p_{\text{Holm}} = 1.9 \times 10^{-6}$), so we present traversal as a bounded complementary channel rather than a headline gain.

Table 15: Provider failures recorded during the collection session (window 2026-08-12 12:09:28 to 2026-08-12 13:47:18).

Signal	Count
Semantic Scholar HTTP 429	78
HTTP 429 (all providers)	78
J-STAGE parse failures	120
Citation-graph failures	0
Translation calls logged	60

16 Provider failures and operational notes

The J-STAGE parse failures were traced to a third-party client that emitted percent-encoded URLs; the channel was replaced by a pure-Python backend and returns real results from v0.4.1 onward. Semantic Scholar anonymous access is aggressively rate-limited from shared/data-centre IPs, which is why an API key is recommended for batch use.

17 Cost measurements

Measurement	Value	Artifact
Full pipeline (composed), median	28.65 s/variant	`timing_43variants.json`
Full pipeline (composed), p90	44.61 s/variant	`timing_43variants.json`
Download rate (end-to-end)	3.99 PDFs/min	`download_throughput.json`
Candidate→PDF conversion	9.6% (18/187)	`download_throughput.json`
Seconds per PDF	15.06	`download_throughput.json`
Peak RSS	85.7 MB	`download_throughput.json`

The timing figure is a composition of the translation-phase run and the citation phase from the same 43-variant session, not a fresh end-to-end run, so that it stays tied to the frozen benchmark lists. The download measurement covers four variants and is a small-sample estimate; it also showed that the download layer returns nothing without `LIT_UNPAYWALL_EMAIL`, because every DOI→open-access route is then skipped.

18 Reproducibility

```
pip install lit-acquisition # lite edition (no LLM calls)
pip install "lit-acquisition[agent]" # + translation, gate, rerank

python benchmarks/run_benchmark.py --collect --limit 100 --cg-budget 20
python benchmarks/make_paper_data.py # regenerates paper numbers
python benchmarks/measure_download_and_memory.py --variants 4 --limit 8
python benchmarks/compose_timing.py
python paper/appnote/make_figure1.py
python paper/appnote/make_supplementary_tables.py
python paper/appnote/check_numbers.py # verifies the manuscript
```

Environment of the recorded run: Python 3.12.3, Linux. Provider API keys were read from the environment; the Unpaywall contact email is required for the DOI→open-access route. Reference-

set and result artifacts live under `benchmarks/`; they are excluded from the PyPI distribution and must be taken from the repository.

19 Usage examples

Search across all channels (lite edition).

```
import asyncio
from lit_acquisition import online_acquisition_workflow

async def main():
    result = await online_acquisition_workflow({
        "query": "MECP2 Rett syndrome case report",
        "action": "search",
        "limit": 20,
    })
    print(result["summary"]) # one-line outcome
    print(result["diagnostics"]) # per-provider status/latency/counts

asyncio.run(main())
```

Multilingual acquisition with downloads (agent edition).

```
from lit_acquisition import multilingual_acquisition_workflow

result = await multilingual_acquisition_workflow({
    "query": "HFE p.C282Y hereditary haemochromatosis",
    "action": "download",
    "language": "multilingual",
    "limit": 30,
    "download_path": "./downloads",
})
```

Citation-graph traversal from a seed DOI.

```
from lit_acquisition import traverse_citation_graph

papers = await traverse_citation_graph(
    seed="10.1038/s41598-022-23012-3", max_depth=1, max_papers=20, direction="both",
)
```